

Hypothesis Testing Worksheet – Research Version

Exercise 7.1 – Related Samples t-test (Container Designs)

Null hypothesis, H_0 : The population mean number of items sold is the same for Design 1 and Design 2.

Alternative hypothesis, H_1 : The population mean number of items sold differs between the two designs.

The paired t-test shows that mean sales are higher for Design 1 than for Design 2.

The estimated difference in means is about 13.2 items per store.

The two-tailed p-value is approximately 0.018, which is below $\alpha = 0.05$.

Conclusion

We reject H_0 at the 5% level. The data provide statistical evidence that Design 1 generates more sales than Design 2, supporting the decision to adopt Design 1 if costs and other constraints are acceptable.

Exercise 7.2 – Independent Samples t-test (Diet A vs Diet B)

Null hypothesis, H_0 : The population mean weight loss is the same for Diet A and Diet B.

Alternative hypothesis, H_1 : The population mean weight losses differ between the two diets.

Results from the equal-variances t-test:

- Mean weight loss: Diet A = 5.341 kg, Diet B = 3.710 kg.
- Difference in means: about 1.631 kg (Diet A – Diet B).
- Two-tailed p-value: about 0.0028, which is below 0.01.

Conclusion

We reject H_0 and conclude that Diet A leads to greater average weight loss than Diet B. The evidence is strong, and the effect is unlikely to be due to chance alone.

Exercise 7.3 – Brand Preferences (Area 1 vs Area 2)

Key percentage preferences in Area 2:

- Brand A: 21.1%
- Brand B: 33.3%
- Other brands: 45.6%

Compared with Area 1 (15.7%, 24.3%, and 60.0%), this indicates that Area 2 has a stronger leaning towards Brand B and less dispersion across “Other” brands. Area 1 shows a more fragmented pattern of preference, with many respondents choosing alternatives beyond the main brands.

Exercise 7.4 – One-Tailed Test (Filtration Agents)

Focus: testing whether Filter Agent 1 is more effective than Filter Agent 2.

Null hypothesis, $H_0: \mu_1 \leq \mu_2$ (Filter Agent 1 is no more effective than Filter Agent 2).

Alternative hypothesis, $H_1: \mu_1 > \mu_2$ (Filter Agent 1 is more effective).

From the one-tailed t-test, Filter Agent 1 performs better on the chosen measure (for example, lower mean impurity), with a p-value around 0.001–0.01, clearly below 0.05.

Conclusion

We reject H_0 in favour of H_1 . The results support the claim that Filter Agent 1 performs better than Filter Agent 2 and would be recommended if the cost is acceptable.

Exercise 7.5 – Independent Samples t-test (Income by Gender)

Null hypothesis, H_0 : The population mean income is the same for males and females.

Alternative hypothesis, H_1 : The population mean income for males exceeds that for females.

The independent samples t-test produces a p-value around 0.0014, which is well below 0.05, with an estimated mean difference in favour of males.

Conclusion

There is evidence to reject H_0 and to conclude that male respondents in this sample have higher mean income than female respondents. This result may also reflect broader socio-economic factors that are not captured in the dataset.

References

Field, A. (2018) Discovering statistics using IBM SPSS Statistics. 5th edn. London: SAGE.

Moore, D.S., Notz, W.I. and Fligner, M.A. (2013) The basic practice of statistics. 6th edn. New York: W.H. Freeman.